

Pranav Dhiran

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PROFILE

Final year undergrad with hands-on experience building LLM-based systems — from transformer pre-training to multi-agent pipelines and open-source LLM-hardware integrations. Actively contributing to open-source and looking to work on large, production-grade codebases where there's real engineering to learn from.

EDUCATION

B.Tech — Electronics & Telecommunication Engineering — SGGS Institute of Engineering & Technology, Nanded

2023 – Present

Minor in IT | Final Year

EXPERIENCE

AI Research Intern — AIISC, University of South Carolina

May 2025 – Present

- Researching neurosymbolic SLM architecture and pre-training pipelines — integrating symbolic reasoning constraints into small language model training for structured knowledge tasks.
- Working on RL-based fine-tuning (GRPO/RLHF) for SLM alignment — reward modeling, policy optimization, and evaluation on neurosymbolic reasoning benchmarks.

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, PyTorch, TensorFlow, Hugging Face Transformers, TRL

APIs & Protocols: REST, gRPC, GraphQL, OpenAI/Gemini/Ollama APIs, FastMCP, XML-RPC, ZMQ

LLM Engineering: Instruction Fine-Tuning, LoRA, PEFT, GRPO, RLHF, INT4/INT8 Quantization, Unsloth, LangChain, LangGraph, ChromaDB, FAISS

Agentic & Infra: Multi-Agent Systems, Tool-Use, Function Calling, MCP Servers, LangSmith, Langfuse, Docker (basic), Kubernetes (basic)

PROJECTS

Small Language Model from Scratch — TinyStories

PyTorch • Hugging Face • AMP • NLP

github.com/Pranav-d33/small_language_model_from_scratch-TinyStories

- Pre-trained a GPT-style autoregressive transformer (6L, 6H, 384-dim) on TinyStories from scratch — custom BPE tokenization, mmap pipelines, AMP mixed precision, gradient accumulation, cosine LR; systematic depth/embedding ablations to analyze scaling behavior.

Medaura — Agentic Pharmacy System

FastAPI • LangChain • ChromaDB • Langfuse • React

aipharmacyproject-blond.vercel.app

- 5-agent autonomous pharmacy system (Ordering, Safety, Forecast, Procurement, UI) with semantic RAG via ChromaDB + sentence-transformers and Langfuse observability tracing every LLM call — live on web, zero human intervention, 4 languages.

GNU Radio MCP Server — LLM-to-SDR Bridge

Python • FastMCP • ZMQ • XML-RPC • GNU Radio

github.com/Pranav-d33/gnuradio-mcp-server

- Open-source provider-agnostic MCP adapter bridging LLMs to live GNU Radio SDR flowgraphs — 13 tools with Pydantic v2 validation, ZMQ context management, stdio/streamable-HTTP transport; async IQ capture with Welch PSD and frequency sweep enabling natural-language control of HackRF/RTL-SDR hardware.

RF Watch — Open-Source RF Spectrum Monitor

Python • GNU Radio • HackRF One • Signal Processing

github.com/Pranav-d33/RFwatch

- Open-source real-time RF spectrum analyzer (HackRF One + GNU Radio) — FFT-based spectral feature extraction with lightweight ML classifier for passive transmitter detection; born from SIH 2025 anti-drone work for ITBP, publicly available for community use.

CERTIFICATIONS

Fine-tuning & RL for LLMs: Intro to Post-training — DeepLearning.AI | AI Agents in LangGraph — DeepLearning.AI | Quantization Fundamentals — Hugging Face | MCP: Build Rich-Context AI Apps — Anthropic

AWARDS & RECOGNITION

- International Finalist (Top 6) — UWA Hack for Impact 2026
- National Finalist — Smart India Hackathon (SIH) 2024 & 2025
- Regional Qualifier — Nxt Wave × OpenAI Buildathon